

M21_20260502

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Load libraries

Set emmeans options

```
emm_options(  
  lmer.df = "kenward-roger",  
  pbkrtest.limit = 10000,  
  disable.pbkrtest = FALSE  
)
```

Set ggplot parameters

```
# Color palette  
my_palette_2 <- c(  
  "#E31A1C", # red  
  "#1F78B4" # blue  
)  
  
# Theme  
my_theme <- theme_classic(base_size = 14) +  
  theme(  
    legend.position = "bottom",  
    legend.direction = "horizontal",  
    legend.box = "horizontal"  
  )  
  
# Common guides  
my_guides <- guides(  
  colour = guide_legend(nrow = 1),  
  fill = guide_legend(nrow = 1)  
)
```

Define 2-way interaction table function

1 Load Files

```
# Load data: rt words
rt_w <- read_csv("rt_words_frq.csv") |>
  filter(correct == 1) |>
  mutate(
    SubjID = factor(SubjID),
    Item = factor(target),
    logRT = log(response_time),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1),
    zLogBF = as.numeric(scale(LogBF)),
    zLogFS = as.numeric(scale(LogFS))
  )

# Load data: rt nonwords
rt_nw <- read_csv("rt_nwords_frq.csv") |>
  filter(correct == 1) |>
  mutate(
    SubjID = factor(SubjID),
    ItemID = factor(ItemID),
    complexity = factor(complexity,
      levels = c("SIMP", "COMP"),
      labels = c("Simple", "Complex")),
    logRT = log(response_time),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1),
    zLogBF = as.numeric(scale(LogBF)),
    zLogFS = as.numeric(scale(LogFS))
  )

# Load data: erp_fs words
erp_fs_w <- read_csv("erp_fs_dmg_words.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    family_size = factor(family_size,
      levels = c("Small", "Large")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# Load data: erp_fs nonwords
erp_fs_nw <- read_csv("erp_fs_dmg_nonwords.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    family_size = factor(family_size,
      levels = c("Small", "Large")),
    complexity = factor(complexity,
      labels = c("Simple", "Complex")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# Load data: erp_bf words
erp_bf_w <- read_csv("erp_bf_dmg_words.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    base_frequency = factor(base_frequency,
      levels = c("Low", "High")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# Load data: erp_bf nonwords
erp_bf_nw <- read_csv("erp_bf_dmg_nonwords.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    base_frequency = factor(base_frequency,
      levels = c("Low", "High")),
    complexity = factor(complexity,
      labels = c("Simple", "Complex")),
    time_window = factor(time_window),
```

```

    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

```

2 WORDS

2.1 RESPONSE TIME DATA

```

library(lmerTest)

m_rt_words <- lmer(
  logRT ~ zLogBF + zLogFS + OS + LP +
    (1 | SubjID) +
    (1 | Item),
  data = rt_w,
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_rt_words)

```

Model

```

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: logRT ~ zLogBF + zLogFS + OS + LP + (1 | SubjID) + (1 | Item)
|| Data: rt_w
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: -3536.3
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -9.8884 -0.6626 -0.0643  0.5951  5.1184
||
|| Random effects:
||  Groups   Name                Variance Std.Dev.
||  Item      (Intercept)  0.001807  0.04251
||  SubjID    (Intercept)  0.017718  0.13311
||  Residual                    0.029621  0.17211
|| Number of obs: 5847, groups: Item, 100; SubjID, 67
||
|| Fixed effects:
||              Estimate Std. Error    df t value Pr(>|t|)
|| (Intercept)  6.390069   0.016965  72.434536 376.655 < 2e-16 ***
|| zLogBF       -0.017184   0.004696  100.974079  -3.659 0.000404 ***
|| zLogFS       -0.012858   0.004865   93.591020  -2.643 0.009629 **
|| OS           -0.034712   0.018734  63.842814  -1.853 0.068518 .
|| LP           0.011591   0.012386  63.973484   0.936 0.352893
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||      (Intr) zLogBF zLogFS OS
|| zLogBF    0.009
|| zLogFS    0.002 -0.105
|| OS        0.012 -0.001  0.001
|| LP        0.008  0.000  0.001  0.008
||
|| # Effect sizes for individual terms
eta_squared(m_rt_words, partial = TRUE)

|| # Effect Size for ANOVA (Type III)
||
|| Parameter | Eta2 (partial) |      95% CI
|| -----|-----|-----
|| zLogBF    |          0.12 | [0.04, 1.00]
|| zLogFS    |          0.07 | [0.01, 1.00]
|| OS        |          0.05 | [0.00, 1.00]
|| LP        |          0.01 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].
||
|| # Model R²
r2_nakagawa(m_rt_words)

|| # R2 for Mixed Models
||
|| Conditional R2: 0.417
|| Marginal R2: 0.033

```

2.2 ERP DATA

2.2.1 N250 DATA: Base Frequency

```
contrasts(erp_bf_w$base_frequency) <- c(-0.5, 0.5)

m_w_N250_bf <- lmer(
  value ~ base_frequency * OS + LP +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_bf_w, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N250_bf)
```

Model

```
|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ base_frequency * OS + LP + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_bf_w, time_window == "N250")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 18999.1
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -7.4496 -0.3820 -0.0038  0.3905 14.0821
||
|| Random effects:
|| Groups          Name                Variance Std.Dev.
|| SubjID:chlabel (Intercept) 19.723    4.441
|| SubjID          (Intercept) 12.044    3.470
|| Residual                        8.095    2.845
|| Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error    df t value Pr(>|t|)
|| (Intercept)    -1.5059     0.4650   56.9998  -3.239  0.00201 **
|| base_frequency1 -0.3183     0.1001 1617.9999  -3.181  0.00150 **
|| OS              0.5341     0.5200   56.9998   1.027  0.30874
|| LP             -0.5356     0.3781   56.9998  -1.417  0.16203
|| base_frequency1:OS -0.4549     0.1120 1617.9999  -4.062 5.09e-05 ***
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||              (Intr) bs_fr1 OS      LP
|| bas_frnc1    0.000
|| OS           -0.046 0.000
|| LP           0.038 0.000 0.021
|| bs_frnc1:OS  0.000 -0.047 0.000 0.000
||
|| # Effect sizes for individual terms
|| eta_squared(m_w_N250_bf, partial = TRUE)
```

```
|| # Effect Size for ANOVA (Type III)
||
|| Parameter          | Eta2 (partial) |      95% CI
|| -----|-----|
|| base_frequency     |      6.21e-03 | [0.00, 1.00]
|| OS                 |      0.02 | [0.00, 1.00]
|| LP                 |      0.03 | [0.00, 1.00]
|| base_frequency:OS |      0.01 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].
||
|| # Model R²
|| r2_nakagawa(m_w_N250_bf)
```

```
|| # R2 for Mixed Models
||
|| Conditional R2: 0.801
|| Marginal R2: 0.018
```

```
emmeans(m_w_N250_bf, ~ base_frequency)
```

Means and contrasts

```
|| base_frequency emmean    SE    df lower.CL upper.CL
```

```

|| Low          -1.29 0.467 58.3    -2.22  -0.355
|| High         -1.63 0.467 58.3    -2.56  -0.692
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95
emm_bf_OS <- emmeans(m_w_N250_bf,
  ~ base_frequency | OS,
  at = list(OS = c(-1, 0, 1)))

emm_part <- summary(emm_bf_OS) |>
  as.data.frame() |>
  transmute(OS, Effect = paste("EMM:", base_frequency), Estimate = emmean, SE, df,
    lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)

contrast_part <- pairs(emm_bf_OS) |>
  summary() |>
  as.data.frame() |>
  transmute(OS, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
    lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

(emm_contrast_table <- bind_rows(emm_part, contrast_part) |>
  arrange(OS, Effect)) |>
  mutate(
    Estimate = round(Estimate, 2),
    SE = round(SE, 2),
    df = round(df, 1),
    lower.CL = round(lower.CL, 2),
    upper.CL = round(upper.CL, 2),
    t.ratio = round(t.ratio, 2),
    p.value = round(p.value, 3)
  )
)

|| OS          Effect Estimate SE df lower.CL upper.CL t.ratio p.value
|| 1 -1 Contrast: Low - High -0.14 0.15 1618.0 NA NA -0.89 0.374
|| 2 -1 EMM: High -1.95 0.72 58.3 -3.38 -0.51 NA NA
|| 3 -1 EMM: Low -2.08 0.72 58.3 -3.52 -0.65 NA NA
|| 4 0 Contrast: Low - High 0.32 0.10 1618.0 NA NA 3.18 0.001
|| 5 0 EMM: High -1.64 0.47 58.3 -2.57 -0.70 NA NA
|| 6 0 EMM: Low -1.32 0.47 58.3 -2.26 -0.39 NA NA
|| 7 1 Contrast: Low - High 0.77 0.15 1618.0 NA NA 5.27 0.000
|| 8 1 EMM: High -1.33 0.68 58.3 -2.70 0.04 NA NA
|| 9 1 EMM: Low -0.56 0.68 58.3 -1.93 0.81 NA NA
# kable(emm_contrast_table, digits = 3)

```

Plots

2.2.2 N250 DATA: Family Size

```

contrasts(erp_fs_w$family_size) <- c(-0.5, 0.5)

m_w_N250_fs <- lmer(
  value ~ family_size * OS + LP +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_fs_w, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N250_fs)

```

Model

```

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ family_size * OS + LP + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_fs_w, time_window == "N250")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 15026.8
||
|| Scaled residuals:
||    Min       1Q   Median       3Q      Max
|| -8.4669 -0.4045 -0.0218  0.3824 12.9276
||
|| Random effects:
|| Groups      Name                Variance Std.Dev.
|| SubjID:chlabel (Intercept)  4.625     2.151
|| SubjID       (Intercept)  2.918     1.708
|| Residual                    2.754     1.659

```

```

|| Number of obs: 3240, groups:  SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||               Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)    -0.75979    0.22921   57.00000   -3.315   0.0016 **
|| family_size1    0.32357    0.05837  1617.99999    5.543 3.46e-08 ***
|| OS              0.26085    0.25635   57.00000    1.018   0.3132
|| LP             -0.27860    0.18636   57.00000   -1.495   0.1404
|| family_size1:OS  0.11120    0.06532  1617.99999    1.702   0.0889 .
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||               (Intr) fmly_1 OS      LP
|| family_size1  0.000
|| OS           -0.046  0.000
|| LP            0.038  0.000  0.021
|| fmly_sz1:OS  0.000 -0.047  0.000  0.000

```

```

# Effect sizes for individual terms
eta_squared(m_w_N250_fs, partial = TRUE)

```

```

|| # Effect Size for ANOVA (Type III)
||
|| Parameter      | Eta2 (partial) |      95% CI
|| -----
|| family_size    |          0.02 | [0.01, 1.00]
|| OS              |          0.02 | [0.00, 1.00]
|| LP              |          0.04 | [0.00, 1.00]
|| family_size:OS |       1.79e-03 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].

```

```

# Model R²
r2_nakagawa(m_w_N250_fs)

```

```

|| # R2 for Mixed Models
||
|| Conditional R2: 0.738
|| Marginal R2: 0.019

```

```

# family size
emmeans(m_w_N250_fs, ~ family_size)

```

Means and Contrasts

```

|| family_size emmean SE df lower.CL upper.CL
|| Small      -0.900 0.231 58.9   -1.36   -0.438
|| Large      -0.571 0.231 58.9   -1.03   -0.110
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95

```

Plots

2.2.3 N400 DATA: Base Frequency

```

contrasts(erp_bf_w$base_frequency) <- c(-0.5, 0.5)

m_w_N400_bf <- lmer(
  value ~ base_frequency * LP + OS +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_bf_w, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N400_bf)

```

Model

```

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ base_frequency * LP + OS + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_bf_w, time_window == "N400")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 19468.7
||
|| Scaled residuals:

```

```

||      Min      1Q   Median      3Q      Max
|| -11.6094 -0.3522  0.0107   0.3807  10.5439
||
|| Random effects:
|| Groups           Name             Variance Std.Dev.
|| SubjID:chlabel (Intercept) 23.249   4.822
|| SubjID          (Intercept) 18.541   4.306
|| Residual                        9.157   3.026
|| Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)    5.617e-01  5.722e-01  5.700e+01   0.982   0.3304
|| base_frequency1 2.141e-01  1.064e-01  1.618e+03   2.012   0.0444 *
|| LP             4.677e-02  4.652e-01  5.700e+01   0.101   0.9203
|| OS             1.290e+00  6.399e-01  5.700e+01   2.016   0.0485 *
|| base_frequency1:LP 3.444e-01  8.659e-02  1.618e+03   3.977  7.29e-05 ***
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||              (Intr) bs_fr1 LP      OS
|| bas_frncyl1  0.000
|| LP           0.038  0.000
|| OS          -0.046  0.000  0.021
|| bs_frqn1:LP  0.000  0.039  0.000  0.000

```

Effect sizes for individual terms

```

eta_squared(m_w_N400_bf, partial = TRUE)

```

```

|| # Effect Size for ANOVA (Type III)
||
|| Parameter          | Eta2 (partial) |      95% CI
|| -----
|| base_frequency     |      2.50e-03 | [0.00, 1.00]
|| LP                 |      1.77e-04 | [0.00, 1.00]
|| OS                 |           0.07 | [0.00, 1.00]
|| base_frequency:LP  |      9.68e-03 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].

```

Model R²

```

r2_nakagawa(m_w_N400_bf)

```

```

|| # R2 for Mixed Models
||
|| Conditional R2: 0.825
|| Marginal R2: 0.026

```

```

emmeans(m_w_N400_bf, ~ base_frequency)

```

Means and Contrasts

```

|| base_frequency emmean   SE df lower.CL upper.CL
|| Low           0.515 0.574 58  -0.634    1.66
|| High          0.712 0.574 58  -0.436    1.86
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95

```

```

emm_bf_LP <- emmeans(m_w_N400_bf,
  ~ base_frequency | LP,
  at = list(LP = c(-1, 0, 1)))

emm_part <- summary(emm_bf_LP) |>
  as.data.frame() |>
  transmute(LP, Effect = paste("EMM:", base_frequency), Estimate = emmean, SE, df,
    lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)

contrast_part <- pairs(emm_bf_LP) |>
  summary() |>
  as.data.frame() |>
  transmute(LP, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
    lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

(emm_contrast_table <- bind_rows(emm_part, contrast_part) |>
  arrange(LP, Effect)|>
  mutate(
    Estimate = round(Estimate, 2),
    SE       = round(SE, 2),
    df       = round(df, 1),

```

```

lower.CL = round(lower.CL, 2),
upper.CL = round(upper.CL, 2),
t.ratio = round(t.ratio, 2),
p.value = round(p.value, 3)
))

```

```

|| LP          Effect Estimate SE df lower.CL upper.CL t.ratio p.value
|| 1 -1 Contrast: Low - High 0.13 0.13 1618 NA NA 0.97 0.333
|| 2 -1          EMM: High 0.50 0.73 58 -0.95 1.96 NA NA
|| 3 -1          EMM: Low 0.63 0.73 58 -0.82 2.09 NA NA
|| 4 0 Contrast: Low - High -0.21 0.11 1618 NA NA -2.01 0.044
|| 5 0          EMM: High 0.72 0.57 58 -0.43 1.87 NA NA
|| 6 0          EMM: Low 0.51 0.57 58 -0.64 1.66 NA NA
|| 7 1 Contrast: Low - High -0.56 0.14 1618 NA NA -3.99 0.000
|| 8 1          EMM: High 0.94 0.75 58 -0.57 2.45 NA NA
|| 9 1          EMM: Low 0.38 0.75 58 -1.13 1.89 NA NA
# kable(emm_contrast_table, digits = 3)

```

Plots

```

contrasts(erp_fs_w$family_size) <- c(-0.5, 0.5)

m_w_N400_fs <- lmer(
  value ~ family_size * LP + OS +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_fs_w, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N400_fs)

```

Family Size

```

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ family_size * LP + OS + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_fs_w, time_window == "N400")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 15561.8
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -10.9307  -0.3874   0.0081   0.4071   6.2048
||
|| Random effects:
|| Groups          Name          Variance Std.Dev.
|| SubjID:chlabel (Intercept)  5.238     2.289
|| SubjID         (Intercept)  4.575     2.139
|| Residual                        3.304     1.818
|| Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)  2.618e-01  2.842e-01  5.700e+01  0.921  0.3608
|| family_size1  4.522e-01  6.392e-02  1.618e+03  7.074  2.23e-12 ***
|| LP           1.231e-03  2.311e-01  5.700e+01  0.005  0.9958
|| OS           6.771e-01  3.179e-01  5.700e+01  2.130  0.0375 *
|| family_size1:LP -7.114e-02  5.201e-02  1.618e+03 -1.368  0.1716
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||      (Intr) fmly_1 LP      OS
|| family_size1  0.000
|| LP            0.038  0.000
|| OS            -0.046  0.000  0.021
|| fmly_size1:LP  0.000  0.039  0.000  0.000
# Effect sizes for individual terms
eta_squared(m_w_N400_fs, partial = TRUE)

```

```

|| # Effect Size for ANOVA (Type III)
||
|| Parameter | Eta2 (partial) | 95% CI
|| -----|-----|-----
|| family_size | 0.03 | [0.02, 1.00]
|| LP | 4.98e-07 | [0.00, 1.00]
|| OS | 0.07 | [0.00, 1.00]
|| family_size:LP | 1.15e-03 | [0.00, 1.00]

```



```
||
|| - One-sided CIs: upper bound fixed at [1.00].
# Model R2
r2_nakagawa(m_w_N400_fs)
```

```
|| # R2 for Mixed Models
||
|| Conditional R2: 0.756
|| Marginal R2: 0.031
```

```
emmeans(m_w_N400_fs, ~ family_size)
```

Means

```
|| family_size emmean SE df lower.CL upper.CL
|| Small 0.0623 0.286 58.5 -0.5091 0.634
|| Large 0.5179 0.286 58.5 -0.0535 1.089
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95
```

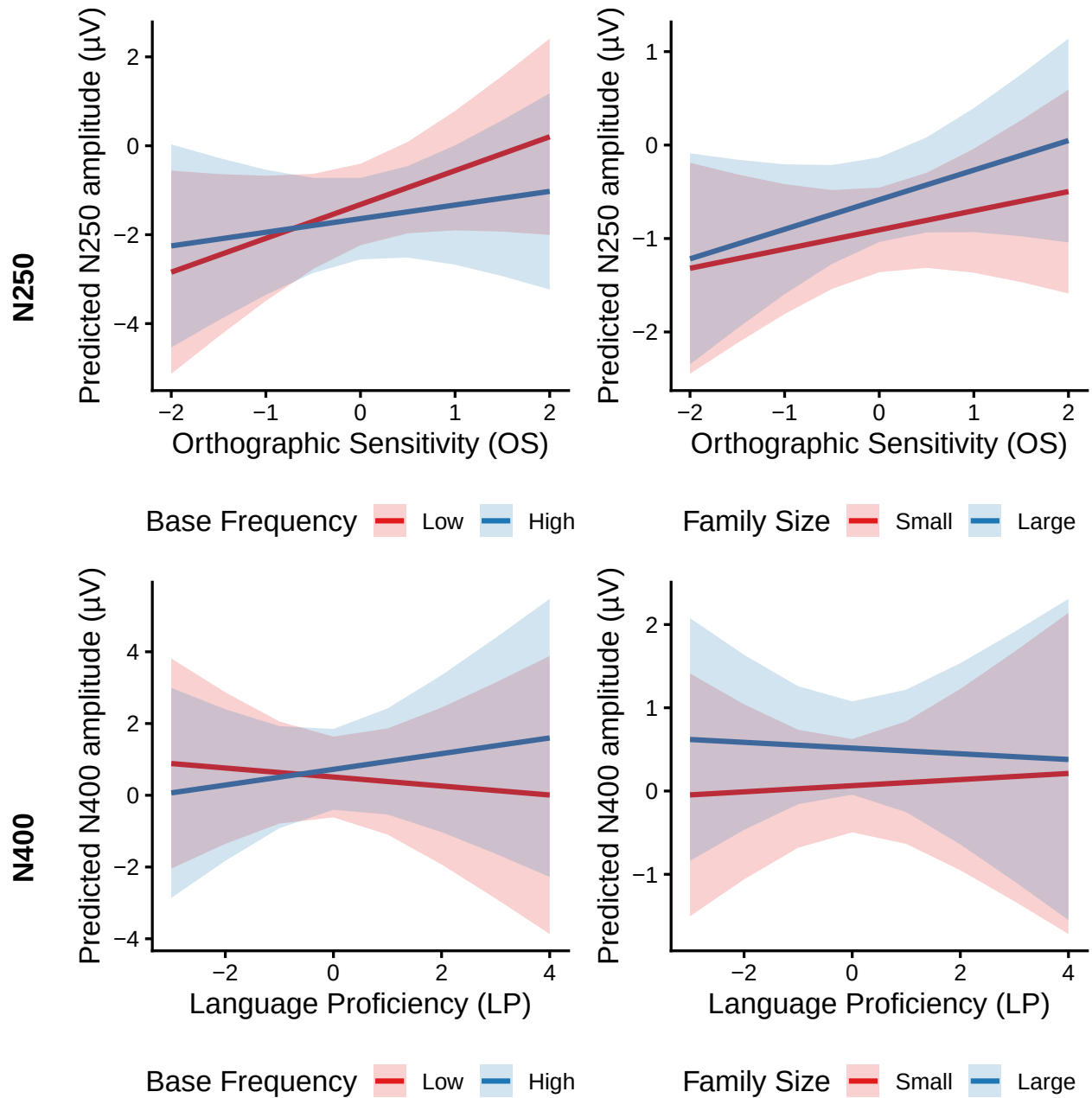
Plots (Interaction Not significant)

2.2.4 N250 WORD DATA: Combined Interaction Plots

```
row1 <- plot_grid(
  BFxOS_n250,
  FSxOS_n250,
  ncol = 2
)

row2 <- plot_grid(
  BFxLP_n400,
  FSxLP_n400,
  ncol = 2
)

plot_grid(
  ggdraw() + draw_label("N250", angle = 90, fontface = "bold"),
  row1,
  ggdraw() + draw_label("N400", angle = 90, fontface = "bold"),
  row2,
  ncol = 2,
  rel_widths = c(0.08, 1)
)
```



3 NON-WORDS

3.1 RESPONSE TIME DATA

```
contrasts(rt_nw$complexity) <- c(-0.5, 0.5)
```

```
m_rt_nw <- lmer(
  logRT ~ complexity + zLogBF + zLogFS + OS + LP +
    (1 + complexity | SubjID) +
    (1 | ItemID),
  data = rt_nw,
  control = lmerControl(optimizer = "bobyqa")
)
```

```
|| Linear mixed model fit by REML ['lmerModLmerTest']
|| Formula: logRT ~ complexity + zLogBF + zLogFS + OS + LP + (1 + complexity | SubjID) + (1 | ItemID)
```

```

|| Data: rt_nw
|| REML criterion at convergence: -4165.782
|| Random effects:
|| Groups Name Std.Dev. Corr
|| ItemID (Intercept) 0.03662
|| SubjID (Intercept) 0.13143
|| complexity1 0.02365 -0.17
|| Residual 0.14641
|| Number of obs: 4607, groups: ItemID, 100; SubjID, 67
|| Fixed Effects:
|| (Intercept) complexity1 zLogBF zLogFS OS LP
|| 6.554486 0.048807 0.015469 -0.006359 -0.041063 0.003398
summary(m_rt_nw)

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: logRT ~ complexity + zLogBF + zLogFS + OS + LP + (1 + complexity | SubjID) + (1 | ItemID)
|| Data: rt_nw
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: -4165.8
||
|| Scaled residuals:
|| Min 1Q Median 3Q Max
|| -4.2605 -0.6625 -0.0227 0.6633 3.8775
||
|| Random effects:
|| Groups Name Variance Std.Dev. Corr
|| ItemID (Intercept) 0.0013408 0.03662
|| SubjID (Intercept) 0.0172747 0.13143
|| complexity1 0.0005594 0.02365 -0.17
|| Residual 0.0214360 0.14641
|| Number of obs: 4607, groups: ItemID, 100; SubjID, 67
||
|| Fixed effects:
|| Estimate Std. Error df t value Pr(>|t|)
|| (Intercept) 6.554486 0.016639 68.832580 393.923 < 2e-16 ***
|| complexity1 0.048807 0.005347 62.675281 9.128 4.07e-13 ***
|| zLogBF 0.015469 0.005080 98.214341 3.045 0.00299 **
|| zLogFS -0.006359 0.005137 96.636227 -1.238 0.21880
|| OS -0.041063 0.018454 62.212180 -2.225 0.02970 *
|| LP 0.003398 0.012228 62.906744 0.278 0.78204
|| ---
|| Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
|| (Intr) cmplx1 zLogBF zLogFS OS
|| complexity1 -0.073
|| zLogBF -0.004 -0.005
|| zLogFS -0.004 0.011 -0.576
|| OS 0.011 -0.007 -0.001 -0.001
|| LP 0.003 -0.015 -0.002 0.000 0.009
# Effect sizes for individual terms
eta_squared(m_rt_nw, partial = TRUE)

|| # Effect Size for ANOVA (Type III)
||
|| Parameter | Eta2 (partial) | 95% CI
|| -----
|| complexity | 0.57 | [0.44, 1.00]
|| zLogBF | 0.09 | [0.02, 1.00]
|| zLogFS | 0.02 | [0.00, 1.00]
|| OS | 0.07 | [0.00, 1.00]
|| LP | 1.23e-03 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].
# Model R^2
r2_nakagawa(m_rt_nw)

|| # R2 for Mixed Models
||
|| Conditional R2: 0.494
|| Marginal R2: 0.050
# Means
# get means on log scale
emm_logrt <- emmeans(m_rt_nw, ~ complexity)
emm_summary <- summary(emm_logrt)

# back transform manually
emm_summary$mean_ms <- exp(emm_summary$emmean)

```

```

emm_summary$lower_ms <- exp(emm_summary$lower.CL)
emm_summary$upper_ms <- exp(emm_summary$upper.CL)
emm_summary

|| complexity emmean      SE    df lower.CL upper.CL mean_ms lower_ms upper_ms
|| Simple      6.529 0.0172 70.03   6.495   6.563   684.6   661.5   708.5
|| Complex     6.578 0.0168 70.15   6.544   6.611   718.9   695.2   743.4
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95
diff_ms <- exp(emm_summary$emmean[emm_summary$complexity == "Complex"]) -
            exp(emm_summary$emmean[emm_summary$complexity == "Simple"])
diff_ms

|| [1] 34.24345

```

3.2 ERP DATA

3.2.1 N250 DATA: Base Frequency as covariate (NOT USED IN WRITE-UP)

```

contrasts(erp_bf_nw$complexity) <- c(-0.5, 0.5)
contrasts(erp_bf_nw$base_frequency) <- c(-0.5, 0.5)

m_nw_N250_bf <- lmer(
  value ~ complexity * OS + complexity * LP + base_frequency +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_bf_nw, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_nw_N250_bf)

```

Model

```

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ complexity * OS + complexity * LP + base_frequency + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_bf_nw, time_window == "N250")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 41527.8
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -12.5220  -0.4882  -0.0036   0.4647  24.5889
||
|| Random effects:
|| Groups          Name                Variance Std.Dev.
|| SubjID:chlabel (Intercept)  10.462    3.235
|| SubjID          (Intercept)   9.777    3.127
|| Residual                        27.474    5.242
|| Number of obs: 6480, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)    -1.4173    0.4175   57.0000  -3.395  0.00126 **
|| complexity1     -0.2243    0.1305  4856.0003  -1.719  0.08566 .
|| OS              0.5743    0.4669   57.0000   1.230  0.22379
|| LP             -0.3205    0.3394   57.0000  -0.944  0.34909
|| base_frequency1 -0.1639    0.1302  4856.0003  -1.258  0.20835
|| complexity1:OS   0.6090    0.1459  4856.0003   4.173  3.05e-05 ***
|| complexity1:LP   0.7230    0.1061  4856.0003   6.816  1.05e-11 ***
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||      (Intr) cmplx1 OS      LP      bs_fr1 cm1:OS
|| complexity1  0.000
|| OS          -0.046  0.000
|| LP           0.038  0.000  0.021
|| bas_frnc1  0.000  0.000  0.000  0.000
|| cmplxty1:OS 0.000 -0.046  0.000  0.000  0.000
|| cmplxty1:LP 0.000  0.038  0.000  0.000  0.000  0.021
||
|| # Effect sizes for individual terms
eta_squared(m_nw_N250_bf, partial = TRUE)

|| # Effect Size for ANOVA (Type III)
||
|| Parameter      | Eta2 (partial) |      95% CI

```

```

|| -----
|| complexity |      6.08e-04 | [0.00, 1.00]
|| OS         |      0.03 | [0.00, 1.00]
|| LP         |      0.02 | [0.00, 1.00]
|| base_frequency | 3.26e-04 | [0.00, 1.00]
|| complexity:OS | 3.57e-03 | [0.00, 1.00]
|| complexity:LP | 9.48e-03 | [0.01, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].

```

```

# Model R2
r2_nakagawa(m_nw_N250_bf)

```

```

|| # R2 for Mixed Models
||
|| Conditional R2: 0.433
|| Marginal R2: 0.015

```

```

## complexity
emmmeans(m_nw_N250_bf, ~ complexity, )

```

Means and Contrasts

```

|| complexity emmean SE df lower.CL upper.CL
|| Simple -1.26 0.422 59.8 -2.10 -0.417
|| Complex -1.49 0.422 59.8 -2.34 -0.651
||
|| Results are averaged over the levels of: base_frequency
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95

```

```

## complexity x LP
emm_bf_LP <- emmeans(m_nw_N250_bf,
  ~ complexity | LP,
  at = list(LP = c(-1, 0, 1)))

emm_part <- summary(emm_bf_LP) |>
  as.data.frame() |>
  transmute(LP, Effect = paste("EMM:", complexity), Estimate = emmean, SE, df,
    lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)

contrast_part <- pairs(emm_bf_LP) |>
  summary() |>
  as.data.frame() |>
  transmute(LP, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
    lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

(emm_contrast_table <- bind_rows(emm_part, contrast_part) |>
  arrange(LP, Effect)|>
  mutate(
    Estimate = round(Estimate, 2),
    SE = round(SE, 2),
    df = round(df, 1),
    lower.CL = round(lower.CL, 2),
    upper.CL = round(upper.CL, 2),
    t.ratio = round(t.ratio, 2),
    p.value = round(p.value, 3)
  ))

```

LP	Effect	Estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
1 -1	Contrast: Simple - Complex	0.92	0.16	4856.0	NA	NA	5.59	0.000
2 -1	EMM: Complex	-1.53	0.53	59.8	-2.60	-0.47	NA	NA
3 -1	EMM: Simple	-0.61	0.53	59.8	-1.68	0.46	NA	NA
4 0	Contrast: Simple - Complex	0.20	0.13	4856.0	NA	NA	1.53	0.127
5 0	EMM: Complex	-1.49	0.42	59.8	-2.34	-0.65	NA	NA
6 0	EMM: Simple	-1.29	0.42	59.8	-2.14	-0.45	NA	NA
7 1	Contrast: Simple - Complex	-0.52	0.17	4856.0	NA	NA	-3.06	0.002
8 1	EMM: Complex	-1.45	0.55	59.8	-2.56	-0.34	NA	NA
9 1	EMM: Simple	-1.98	0.55	59.8	-3.09	-0.87	NA	NA

```

## complexity x OS
emm_bf_OS <- emmeans(m_nw_N250_bf,
  ~ complexity | OS,
  at = list(OS = c(-1, 0, 1)))

emm_part <- summary(emm_bf_OS) |>
  as.data.frame() |>
  transmute(OS, Effect = paste("EMM:", complexity), Estimate = emmean, SE, df,
    lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)

contrast_part <- pairs(emm_bf_OS) |>

```

```
summary() |>
as.data.frame() |>
transmute(OS, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
          lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

(emm_contrast_table <- bind_rows(emm_part, contrast_part) |>
 arrange(OS, Effect)|>
 mutate(
   Estimate = round(Estimate, 2),
   SE       = round(SE, 2),
   df       = round(df, 1),
   lower.CL = round(lower.CL, 2),
   upper.CL = round(upper.CL, 2),
   t.ratio  = round(t.ratio, 2),
   p.value  = round(p.value, 3)
 ))
```

	OS	Effect	Estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
1	-1	Contrast: Simple - Complex	0.87	0.20	4856.0	NA	NA	4.34	0.000
2	-1	EMM: Complex	-2.41	0.65	59.8	-3.71	-1.11	NA	NA
3	-1	EMM: Simple	-1.54	0.65	59.8	-2.84	-0.25	NA	NA
4	0	Contrast: Simple - Complex	0.26	0.13	4856.0	NA	NA	1.99	0.047
5	0	EMM: Complex	-1.53	0.42	59.8	-2.38	-0.69	NA	NA
6	0	EMM: Simple	-1.27	0.42	59.8	-2.12	-0.43	NA	NA
7	1	Contrast: Simple - Complex	-0.35	0.19	4856.0	NA	NA	-1.83	0.067
8	1	EMM: Complex	-0.65	0.62	59.8	-1.89	0.59	NA	NA
9	1	EMM: Simple	-1.00	0.62	59.8	-2.24	0.24	NA	NA

Plots

3.2.2 N250 DATA: Family Size as covariate

```
contrasts(erp_fs_nw$complexity) <- c(-0.5, 0.5)
contrasts(erp_fs_nw$family_size) <- c(-0.5, 0.5)

m_nw_N250_fs <- lmer(
  value ~ complexity * OS + complexity * LP + family_size +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_fs_nw, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_nw_N250_fs)
```

Model

```
|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ complexity * OS + complexity * LP + family_size + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_fs_nw, time_window == "N250")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 29933.2
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -7.7366 -0.5107 -0.0056  0.4938 12.2676
||
|| Random effects:
|| Groups      Name                Variance Std.Dev.
|| SubjID:chlabel (Intercept)  3.109      1.763
|| SubjID       (Intercept)  2.163      1.471
|| Residual                    4.072      2.018
|| Number of obs: 6480, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error    df t value Pr(>|t|)
|| (Intercept)   -0.66509    0.19682  56.99997  -3.379 0.001317 **
|| complexity1   -0.23263    0.05023 4856.00002  -4.631 3.73e-06 ***
|| OS            0.29808    0.22013  56.99997   1.354 0.181051
|| LP            -0.20675    0.16003  56.99997  -1.292 0.201577
|| family_size1  0.18172    0.05014 4856.00002   3.625 0.000292 ***
|| complexity1:OS 0.39199    0.05618 4856.00002   6.978 3.40e-12 ***
|| complexity1:LP 0.35764    0.04084 4856.00002   8.758 < 2e-16 ***
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||      (Intr) cmplx1 OS      LP      fmly_1 cm1:OS
|| complexity1  0.000
```

```

|| OS          -0.046  0.000
|| LP           0.038  0.000  0.021
|| family_size  0.000  0.000  0.000  0.000
|| cmplxty1:OS  0.000 -0.046  0.000  0.000  0.000
|| cmplxty1:LP  0.000  0.038  0.000  0.000  0.000  0.021

```

```

# Effect sizes for individual terms
eta_squared(m_nw_N250_fs, partial = TRUE)

```

```

|| # Effect Size for ANOVA (Type III)
||
|| Parameter      | Eta2 (partial) |      95% CI
|| -----
|| complexity     | 4.40e-03 | [0.00, 1.00]
|| OS             | 0.03 | [0.00, 1.00]
|| LP             | 0.03 | [0.00, 1.00]
|| family_size    | 2.70e-03 | [0.00, 1.00]
|| complexity:OS  | 9.93e-03 | [0.01, 1.00]
|| complexity:LP  | 0.02 | [0.01, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].

```

```

# Model R²
r2_nakagawa(m_nw_N250_fs)

```

```

|| # R2 for Mixed Models
||
|| Conditional R2: 0.575
|| Marginal R2: 0.025

```

```

## complexity
emmmeans(m_nw_N250_fs, ~ complexity)

```

Means and Contrasts

```

|| complexity emmean SE df lower.CL upper.CL
|| Simple -0.526 0.198 58.9 -0.922 -0.130
|| Complex -0.759 0.198 58.9 -1.156 -0.363
||
|| Results are averaged over the levels of: family_size
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95

```

```

## complexity x LP
emm_bf_LP <- emmeans(m_nw_N250_fs,
  ~ complexity | LP,
  at = list(LP = c(-1, 0, 1)))

emm_part <- summary(emm_bf_LP) |>
  as.data.frame() |>
  transmute(LP, Effect = paste("EMM:", complexity), Estimate = emmean, SE, df,
    lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)

contrast_part <- pairs(emm_bf_LP) |>
  summary() |>
  as.data.frame() |>
  transmute(LP, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
    lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

(emm_contrast_table <- bind_rows(emm_part, contrast_part) |>
  arrange(LP, Effect)|>
  mutate(
    Estimate = round(Estimate, 2),
    SE = round(SE, 2),
    df = round(df, 1),
    lower.CL = round(lower.CL, 2),
    upper.CL = round(upper.CL, 2),
    t.ratio = round(t.ratio, 2),
    p.value = round(p.value, 3)
  ))

```

```

|| LP          Effect Estimate SE df lower.CL upper.CL t.ratio p.value
|| 1 -1 Contrast: Simple - Complex 0.57 0.06 4856.0 NA NA 9.05 0.000
|| 2 -1 EMM: Complex -0.73 0.25 58.9 -1.23 -0.23 NA NA
|| 3 -1 EMM: Simple -0.16 0.25 58.9 -0.66 0.34 NA NA
|| 4 0 Contrast: Simple - Complex 0.22 0.05 4856.0 NA NA 4.31 0.000
|| 5 0 EMM: Complex -0.76 0.20 58.9 -1.16 -0.36 NA NA
|| 6 0 EMM: Simple -0.54 0.20 58.9 -0.94 -0.15 NA NA
|| 7 1 Contrast: Simple - Complex -0.14 0.07 4856.0 NA NA -2.14 0.032
|| 8 1 EMM: Complex -0.79 0.26 58.9 -1.31 -0.27 NA NA
|| 9 1 EMM: Simple -0.93 0.26 58.9 -1.45 -0.41 NA NA

```

```
## complexity x OS
emm_bf_OS <- emmeans(m_nw_N250_fs,
  ~ complexity | OS,
  at = list(OS = c(-1, 0, 1)))

emm_part <- summary(emm_bf_OS) |>
  as.data.frame() |>
  transmute(OS, Effect = paste("EMM:", complexity), Estimate = emmean, SE, df,
    lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)

contrast_part <- pairs(emm_bf_OS) |>
  summary() |>
  as.data.frame() |>
  transmute(OS, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
    lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

(emm_contrast_table <- bind_rows(emm_part, contrast_part) |>
  arrange(OS, Effect)|>      # sorts rows in the combined table, first by column "OS", then by "Effect"
  mutate(
    Estimate = round(Estimate, 2),
    SE       = round(SE, 2),
    df       = round(df, 1),
    lower.CL = round(lower.CL, 2),
    upper.CL = round(upper.CL, 2),
    t.ratio  = round(t.ratio, 2),
    p.value  = round(p.value, 3)
  ))
```

	OS	Effect	Estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
1	-1	Contrast: Simple - Complex	0.64	0.08	4856.0	NA	NA	8.33	0.000
2	-1	EMM: Complex	-1.27	0.30	58.9	-1.88	-0.66	NA	NA
3	-1	EMM: Simple	-0.63	0.30	58.9	-1.24	-0.02	NA	NA
4	0	Contrast: Simple - Complex	0.25	0.05	4856.0	NA	NA	4.98	0.000
5	0	EMM: Complex	-0.78	0.20	58.9	-1.18	-0.38	NA	NA
6	0	EMM: Simple	-0.53	0.20	58.9	-0.93	-0.13	NA	NA
7	1	Contrast: Simple - Complex	-0.14	0.07	4856.0	NA	NA	-1.93	0.053
8	1	EMM: Complex	-0.29	0.29	58.9	-0.87	0.30	NA	NA
9	1	EMM: Simple	-0.43	0.29	58.9	-1.01	0.15	NA	NA

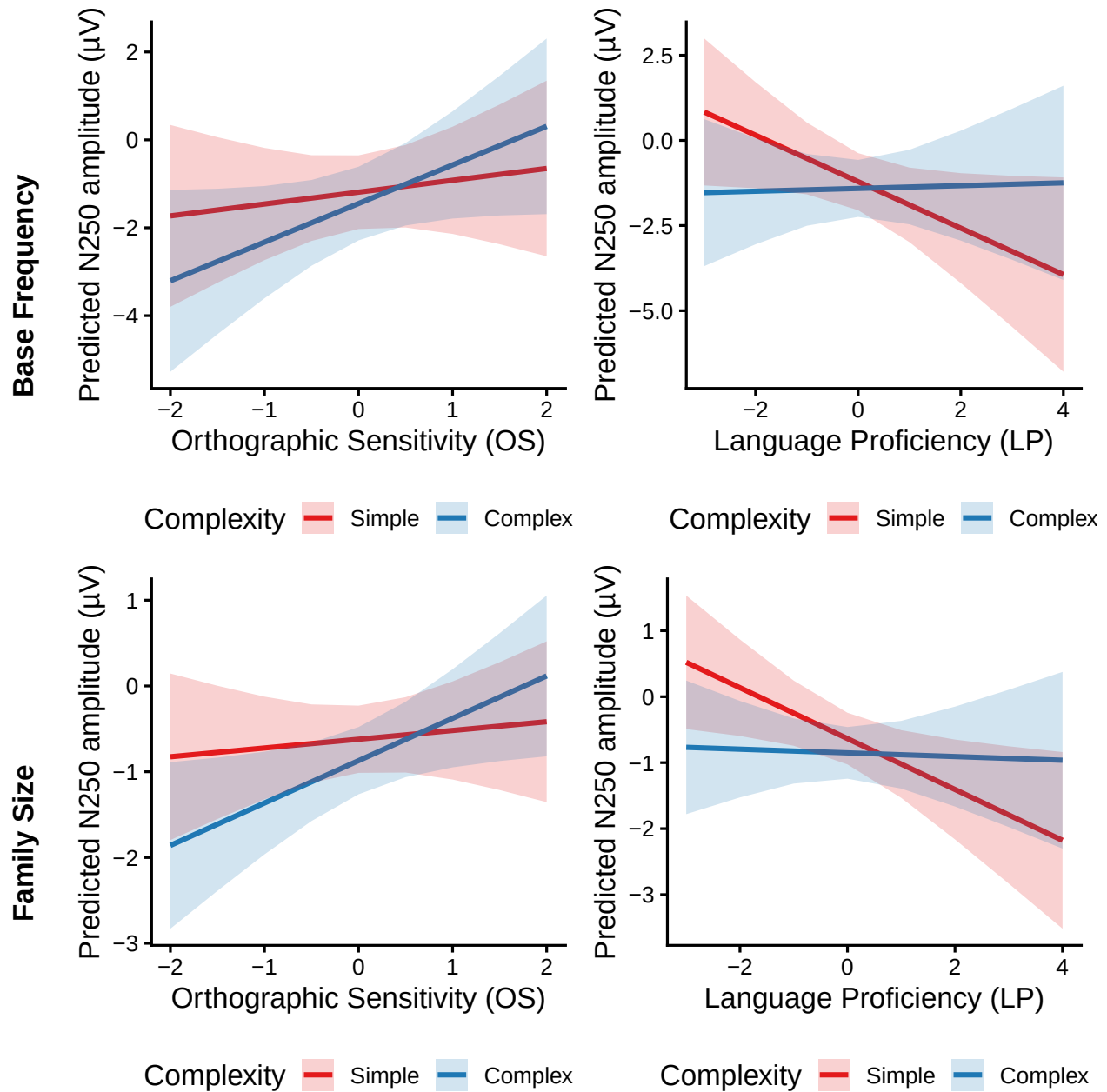
Plots

3.2.3 N250 NONWORD DATA: Combined Interaction Plots

```
row1 <- plot_grid(
  CMP.OS.bf.n250.nw,
  CMP.LP.bf.n250.nw,
  ncol = 2
)

row2 <- plot_grid(
  CMP.OS.fs.n250.nw,
  CMP.LP.fs.n250.nw,
  ncol = 2
)

plot_grid(
  ggdraw() + draw_label("Base Frequency", angle = 90, fontface = "bold"),
  row1,
  ggdraw() + draw_label("Family Size", angle = 90, fontface = "bold"),
  row2,
  ncol = 2,
  rel_widths = c(0.08, 1)
)
```

3.2.4 N400 DATA: Base Frequency (NOT USED IN WRITE-UP)

```
contrasts(erp_bf_nw$complexity) <- c(-0.5, 0.5)
contrasts(erp_bf_nw$base_frequency) <- c(-0.5, 0.5)

m_nw_N400_bf <- lmer(
  value ~ complexity * base_frequency * LP + OS +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_bf_nw, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_nw_N400_bf)
```

Model

```
|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ complexity * base_frequency * LP + OS + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_bf_nw, time_window == "N400")
|| Control: lmerControl(optimizer = "bobyqa")
```

```

||
|| REML criterion at convergence: 44167.9
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -11.9475  -0.3895   0.0032   0.4332  20.9414
||
|| Random effects:
||  Groups             Name             Variance Std.Dev.
||  SubjID:chlabel      (Intercept)    23.71     4.869
||  SubjID              (Intercept)    23.96     4.895
||  Residual                        38.00     6.164
|| Number of obs: 6480, groups:  SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)    -0.84542    0.64909   56.99999  -1.302   0.19799
|| complexity1     -0.44211    0.15327  4854.00000  -2.884   0.00394 **
|| base_frequency1 -0.04368    0.15327  4854.00000  -0.285   0.77566
|| LP              0.23770    0.52775   56.99999   0.450   0.65413
|| OS              0.74117    0.72597   56.99999   1.021   0.31159
|| complexity1:base_frequency1 -0.05967    0.30655  4854.00000  -0.195   0.84566
|| complexity1:LP   0.20562    0.12473  4854.00000   1.649   0.09930
|| base_frequency1:LP 0.11261    0.12473  4854.00000   0.903   0.36663
|| complexity1:base_frequency1:LP -0.17117    0.24945  4854.00000  -0.686   0.49264
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||              (Intr) cmplx1 bs_fr1 LP      OS      cm1:_1 cm1:LP b_1:LP
|| complexity1  0.000
|| bas_frqn1    0.000 0.000
|| LP           0.038 0.000 0.000
|| OS          -0.046 0.000 0.000 0.021
|| cmplxty1:_1  0.000 0.000 0.000 0.000 0.000
|| cmplxty1:LP  0.000 0.039 0.000 0.000 0.000 0.000
|| bs_frqn1:LP  0.000 0.000 0.039 0.000 0.000 0.000 0.000
|| cmpl1:_1:LP  0.000 0.000 0.000 0.000 0.000 0.039 0.000 0.000
||
|| # Effect sizes for individual terms
|| eta_squared(m_nw_N400_bf, partial = TRUE)
||
|| # Effect Size for ANOVA (Type III)
||
|| Parameter | Eta2 (partial) | 95% CI
|| -----|-----|-----|
|| complexity | 1.71e-03 | [0.00, 1.00]
|| base_frequency | 1.67e-05 | [0.00, 1.00]
|| LP | 3.55e-03 | [0.00, 1.00]
|| OS | 0.02 | [0.00, 1.00]
|| complexity:base_frequency | 7.81e-06 | [0.00, 1.00]
|| complexity:LP | 5.60e-04 | [0.00, 1.00]
|| base_frequency:LP | 1.68e-04 | [0.00, 1.00]
|| complexity:base_frequency:LP | 9.70e-05 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].
||
|| # Model R²
|| r2_nakagawa(m_nw_N400_bf)
||
|| # R2 for Mixed Models
||
|| Conditional R2: 0.559
|| Marginal R2: 0.007

```

3.2.5 400 DATA: Family Size

```

contrasts(erp_fs_nw$complexity) <- c(-0.5, 0.5)
contrasts(erp_fs_nw$family_size) <- c(-0.5, 0.5)

m_nw_N400_fs <- lmer(
  value ~ complexity * family_size * LP + OS +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_fs_nw, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_nw_N400_fs)

```

Model

```

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ complexity * family_size * LP + OS + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_fs_nw, time_window == "N400")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 32979.6
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -13.9873  -0.4837  -0.0118   0.4655  14.1595
||
|| Random effects:
||      Groups             Name             Variance Std.Dev.
|| SubjID:chlabel (Intercept)  6.389       2.528
|| SubjID         (Intercept)  5.568       2.360
|| Residual                        6.114       2.473
|| Number of obs: 6480, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)    -0.45163    0.31311   57.00002  -1.442 0.154666
|| complexity1     -0.45956    0.06148 4853.99999  -7.475 9.11e-14 ***
|| family_size1     0.08743    0.06148 4853.99999   1.422 0.155086
|| LP              0.01139    0.25458   57.00002   0.045 0.964463
|| OS              0.38123    0.35019   57.00002   1.089 0.280904
|| complexity1:family_size1 -0.44008    0.12296 4853.99999  -3.579 0.000348 ***
|| complexity1:LP    0.25043    0.05003 4853.99999   5.006 5.77e-07 ***
|| family_size1:LP    0.12120    0.05003 4853.99999   2.423 0.015450 *
|| complexity1:family_size1:LP 0.22717    0.10006 4853.99999   2.270 0.023230 *
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||      (Intr) cmplx1 fmly_1 LP      OS      cm1:_1 cm1:LP f_1:LP
|| complexity1  0.000
|| family_size1 0.000 0.000
|| LP           0.038 0.000 0.000
|| OS          -0.046 0.000 0.000 0.021
|| cmplxty1:_1 0.000 0.000 0.000 0.000 0.000
|| cmplxty1:LP 0.000 0.039 0.000 0.000 0.000 0.000
|| fmly_sz1:LP 0.000 0.000 0.039 0.000 0.000 0.000 0.000
|| cmpl1:_1:LP 0.000 0.000 0.000 0.000 0.000 0.039 0.000 0.000
||
|| # Effect sizes for individual terms
|| eta_squared(m_nw_N400_fs, partial = TRUE)
||
|| # Effect Size for ANOVA (Type III)
||
|| Parameter | Eta2 (partial) | 95% CI
|| -----|-----|-----|
|| complexity | 0.01 | [0.01, 1.00]
|| family_size | 4.16e-04 | [0.00, 1.00]
|| LP | 3.51e-05 | [0.00, 1.00]
|| OS | 0.02 | [0.00, 1.00]
|| complexity:family_size | 2.63e-03 | [0.00, 1.00]
|| complexity:LP | 5.14e-03 | [0.00, 1.00]
|| family_size:LP | 1.21e-03 | [0.00, 1.00]
|| complexity:family_size:LP | 1.06e-03 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].
||
|| # Model R²
|| r2_nakagawa(m_nw_N400_fs)
||
|| # R2 for Mixed Models
||
|| Conditional R2: 0.666
|| Marginal R2: 0.012
||
|| emm_3way <- emmeans(
||   m_nw_N400_fs,
||   ~ complexity * family_size | LP,
||   at = list(LP = c(-1, 0, 1))
|| )
||
|| compact_3way_table(
||   emm_3way,
||   by_var = LP,
||   condition_prefix = "LP",
||   factor1 = complexity,

```

```
factor2 = family_size
)
```

Means and Contrasts: Complexity × Family Size × LP interaction

	Comparison	Condition	Estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
1	Complex Large	LP = -1	-0.99	0.40	60.4	-1.79	-0.18	NA	NA
2	Complex Small	LP = -1	-0.62	0.40	60.4	-1.42	0.18	NA	NA
3	Simple Large	LP = -1	0.06	0.40	60.4	-0.74	0.86	NA	NA
4	Simple Small	LP = -1	-0.24	0.40	60.4	-1.04	0.56	NA	NA
5	Int: Simple - Complex across	Small - Large	LP = -1	-0.67	0.16	4854.0	NA	NA	-4.29 0.000
6	Small - Large	LP = -1 complexity = Complex	0.37	0.11	4854.0	NA	NA	3.34	0.001
7	Small - Large	LP = -1 complexity = Simple	-0.30	0.11	4854.0	NA	NA	-2.73	0.006
8	Simple - Complex	LP = -1 family_size = Large	1.04	0.11	4854.0	NA	NA	9.49	0.000
9	Simple - Complex	LP = -1 family_size = Small	0.38	0.11	4854.0	NA	NA	3.42	0.001
10	Complex Large	LP = 0	-0.73	0.32	60.4	-1.37	-0.10	NA	NA
11	Complex Small	LP = 0	-0.60	0.32	60.4	-1.23	0.04	NA	NA
12	Simple Large	LP = 0	-0.05	0.32	60.4	-0.69	0.58	NA	NA
13	Simple Small	LP = 0	-0.36	0.32	60.4	-0.99	0.27	NA	NA
14	Int: Simple - Complex across	Small - Large	LP = 0	-0.44	0.12	4854.0	NA	NA	-3.58 0.000
15	Small - Large	LP = 0 complexity = Complex	0.13	0.09	4854.0	NA	NA	1.53	0.127
16	Small - Large	LP = 0 complexity = Simple	-0.31	0.09	4854.0	NA	NA	-3.54	0.000
17	Simple - Complex	LP = 0 family_size = Large	0.68	0.09	4854.0	NA	NA	7.82	0.000
18	Simple - Complex	LP = 0 family_size = Small	0.24	0.09	4854.0	NA	NA	2.75	0.006
19	Complex Large	LP = 1	-0.48	0.42	60.3	-1.31	0.36	NA	NA
20	Complex Small	LP = 1	-0.58	0.42	60.3	-1.41	0.25	NA	NA
21	Simple Large	LP = 1	-0.16	0.42	60.3	-1.00	0.67	NA	NA
22	Simple Small	LP = 1	-0.48	0.42	60.3	-1.31	0.36	NA	NA
23	Int: Simple - Complex across	Small - Large	LP = 1	-0.21	0.16	4854.0	NA	NA	-1.32 0.188
24	Small - Large	LP = 1 complexity = Complex	-0.10	0.11	4854.0	NA	NA	-0.89	0.371
25	Small - Large	LP = 1 complexity = Simple	-0.32	0.11	4854.0	NA	NA	-2.76	0.006
26	Simple - Complex	LP = 1 family_size = Large	0.32	0.11	4854.0	NA	NA	2.76	0.006
27	Simple - Complex	LP = 1 family_size = Small	0.10	0.11	4854.0	NA	NA	0.90	0.369

```
emm_complexity_FS_N400 <- emmeans(
  m_nw_N400_fs,
  ~ complexity | family_size)

compact_emm_table(
  emm_complexity_FS_N400,
  by_var = family_size
)
```

Complexity × Family Size interaction averaged across LP

#	Comparison	Condition	Estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
1	Complex Large		-0.74	0.32	60.4	-1.38	-0.11	NA	NA
2	Simple Large		-0.05	0.32	60.4	-0.68	0.59	NA	NA
3	Simple - Complex Large		0.7	0.09	4854	NA	NA	8.02	0
4	Complex Small		-0.6	0.32	60.4	-1.23	0.03	NA	NA
5	Simple Small		-0.35	0.32	60.4	-0.99	0.28	NA	NA
6	Simple - Complex Small		0.25	0.09	4854	NA	NA	2.83	0.005

```
emm_complexity_CMP_N400 <- emmeans(
  m_nw_N400_fs,
  ~ family_size | complexity)

compact_emm_table(
  emm_complexity_CMP_N400,
  by_var = complexity
)
```

#	Comparison	Condition	Estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
1	Large Complex		-0.74	0.32	60.4	-1.38	-0.11	NA	NA
2	Small Complex		-0.6	0.32	60.4	-1.23	0.03	NA	NA
3	Small - Large Complex		0.14	0.09	4854	NA	NA	1.66	0.098
4	Large Simple		-0.05	0.32	60.4	-0.68	0.59	NA	NA
5	Small Simple		-0.35	0.32	60.4	-0.99	0.28	NA	NA
6	Small - Large Simple		-0.31	0.09	4854	NA	NA	-3.53	0

```
emm_complexity_LP_N400 <- emmeans(
  m_nw_N400_fs,
  ~ complexity | LP,
  at = list(LP = c(-1, 0, 1))
)
```

```
compact_emm_table(
  emm_complexity_LP_N400,
  by_var = LP,
  condition_prefix = "LP"
)
```

Complexity × LP interaction averaged over Family Size:

```
|| # A tibble: 9 x 9
|| Comparison      Condition Estimate      SE      df lower.CL upper.CL t.ratio p.value
|| <chr>            <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Complex        LP = -1      -0.8  0.4   58.1  -1.6  -0.01  NA      NA
|| 2 Simple          LP = -1      -0.09 0.4   58.1  -0.89  0.7   NA      NA
|| 3 Simple - Complex LP = -1      0.71 0.08 4854   NA     NA     9.13    0
|| 4 Complex        LP = 0       -0.67 0.31  58.1  -1.29 -0.04  NA      NA
|| 5 Simple          LP = 0       -0.21 0.31  58.1  -0.84  0.42  NA      NA
|| 6 Simple - Complex LP = 0       0.46 0.06 4854   NA     NA     7.47    0
|| 7 Complex        LP = 1      -0.53 0.41  58.1  -1.36  0.3   NA      NA
|| 8 Simple          LP = 1      -0.32 0.41  58.1  -1.15  0.51  NA      NA
|| 9 Simple - Complex LP = 1       0.21 0.08 4854   NA     NA     2.59    0.01
```

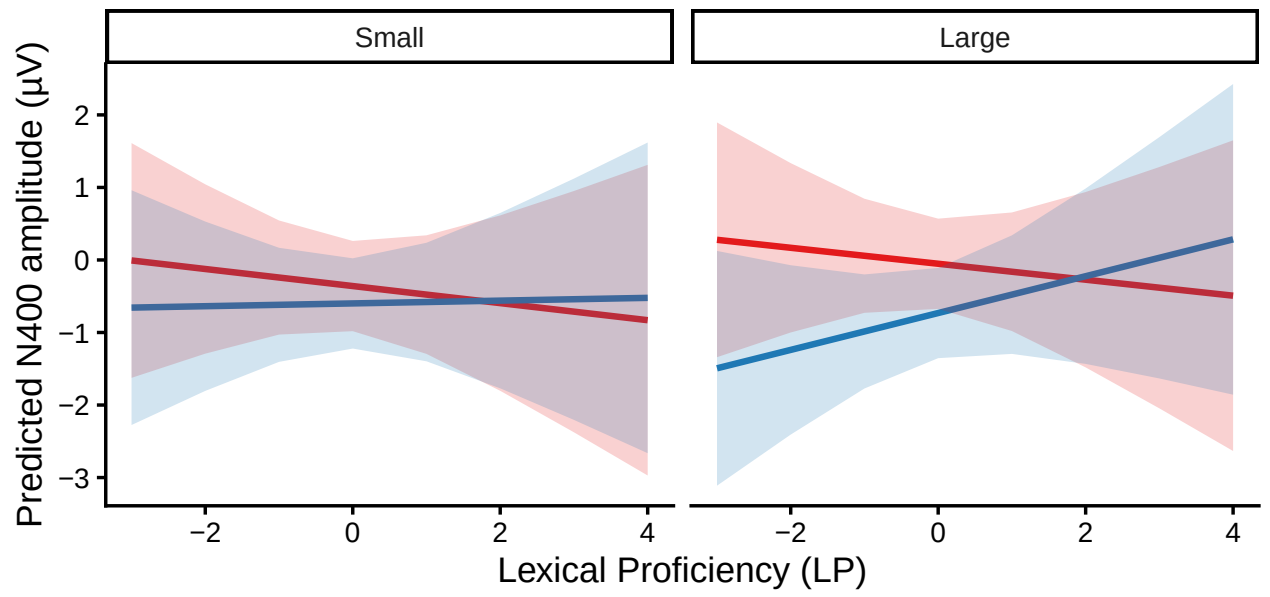
```
emm_FS_LP_N400 <- emmeans(
  m_nw_N400_fs,
  ~ family_size | LP,
  at = list(LP = c(-1, 0, 1))
)

compact_emm_table(
  emm_FS_LP_N400,
  by_var = LP,
  condition_prefix = "LP"
)
```

FS × LP interaction averaged over Complexity:

```
|| # A tibble: 9 x 9
|| Comparison      Condition Estimate      SE      df lower.CL upper.CL t.ratio p.value
|| <chr>            <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Large          LP = -1      -0.46 0.4   58.1  -1.26  0.33  NA      NA
|| 2 Small          LP = -1      -0.43 0.4   58.1  -1.23  0.37  NA      NA
|| 3 Small - Large  LP = -1      0.03 0.08 4854   NA     NA     0.43    0.664
|| 4 Large          LP = 0       -0.39 0.31  58.1  -1.02  0.24  NA      NA
|| 5 Small          LP = 0       -0.48 0.31  58.1  -1.11  0.15  NA      NA
|| 6 Small - Large  LP = 0       -0.09 0.06 4854   NA     NA    -1.42    0.155
|| 7 Large          LP = 1      -0.32 0.41  58.1  -1.15  0.51  NA      NA
|| 8 Small          LP = 1      -0.53 0.41  58.1  -1.36  0.3   NA      NA
|| 9 Small - Large  LP = 1      -0.21 0.08 4854   NA     NA    -2.58    0.01
```

```
plot_grid(nw.n400.3way, ncol = 1)
```



Plots

Complexity Simple Complex